FABRICATION PROCESSES IN THE IC MANUFACTURING

2023EV093

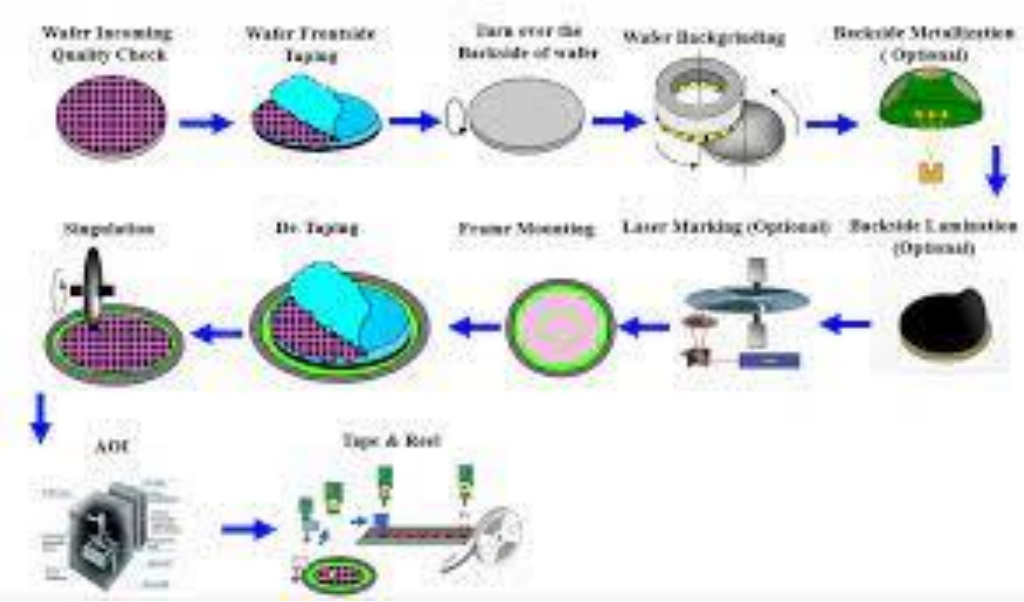
DEPARTMENT OF ECE

1. IC FABRICATION PROCESS:

Integrated circuits are a collection of electronic circuits that are arranged on a tiny electronic chip. The fabrication method is used to create a wide range of devices such as transistors, MOSFETs (Metal Oxide Semiconductor Field Effect Transistors), microcontrollers, computer processors, and so on. Silicon and germanium are the most commonly used materials in the fabrication of semiconductors due to their stable structures. Wood, thermoplastics, resins, silver, aluminium, magnesium, copper, and carbon steel are also popular in the fabrication industry.

Let's start with fabrication and the requirements it has in the semiconductor industry.

Fabrication is the process of creating an industrial product. It can also be defined as a collection of techniques used in the manufacture of electronic goods. As an example, consider silicon semiconductor chips.



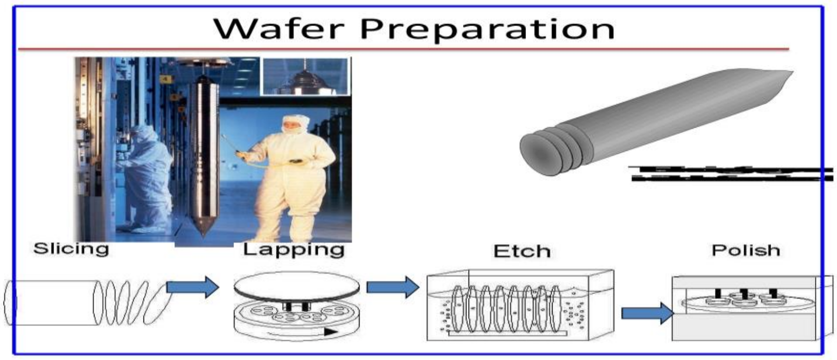
1.1TYPES OF IC FABRICATION:

The fundamental techniques used in fabricating integrated circuits:

- **WAFER PREPARATION**
- **OXIDATION**
- **DIFFUSION**
- **ION IMPLANTATION**
- **CHEMICAL-VAPOR DEPOSITION**
- **PHOTOLITHOGRAPHY**
- **METALIZATION**
- **PACKAGING**

1.1.1 WAFER PREPARATION:

A wafer is a thin piece of material that is used to make different types of transistors and integrated circuits. A wafer serves as the foundation for such devices. A wafer is a semiconductor, specifically crystalline silicon. Wafers are made from extremely pure silicon crystals. A boule is a method of extracting pure metal from a melt. To make the material n-type or p-type, more impurities are added to it precisely while it is still molten..



1.1.2 OXIDATION:

Oxidation is the process of introducing oxygen. When silicon and oxygen interact in a semiconductor, silicon dioxide is formed. Oxidation occurs in furnaces at temperatures as high as 1250 degrees

Celsius. There are two types of oxidations: wet oxidation and dry oxidation. Both procedures are widely used and each has advantages and disadvantages. Wet oxidation is faster than dry oxidation because it has stronger electrical properties.

Wet oxidation is also referred to as "steam". Both types of oxidation have excellent electrical insulating properties. Only areas not covered in SiO2 may be treated with dopants. An oxide layer on a substrate is depicted below:

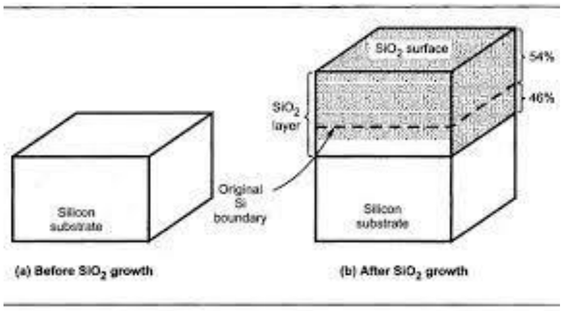


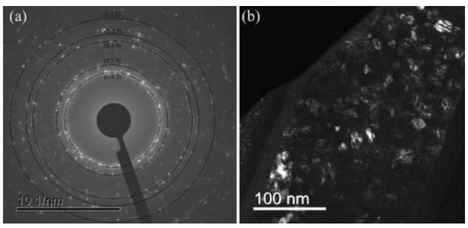
Fig. 1.4 Thermal Oxidation

1.1.3 DIFFUSION:

The addition of impurity atoms from a high concentration region to a low concentration region is referred to as diffusion. Dopants, or impurity atoms, change the resistivity of silicon, a semiconductor material. The diffusion process is greatly influenced by temperature. It is carried out in high-temperature furnaces with temperatures ranging from 1000 to 1200 degrees Celsius. The depth and width of the impurities are affected by the temperature range and timings. The high doping concentration improves a metal's conductivity.

1.1.4 ION IMPLANTATION:

Ion implantation is the process of accelerating ions from the element to the solid target. The accelerated ions can change the properties of the solid. Ions are used with high energy and at a low temperature. Because of the low temperature, it may operate at room temperature. The extra energy, however, has the potential to harm the solid's crystal structure..

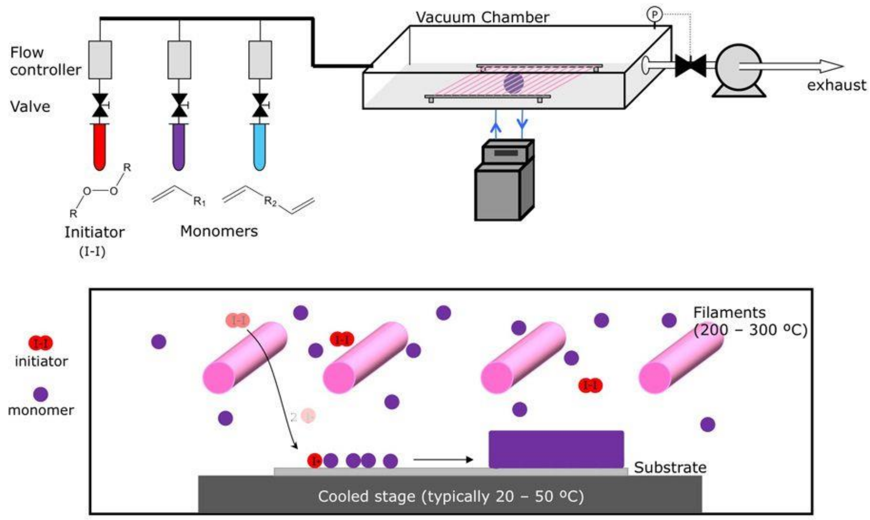


Ion implantation introduces impurity atoms into the crystal (semiconductor material). When the speeding atom collides with the surface of the crystal, it becomes embedded. Energy and accelerating field voltage determine the depth of ion penetration.

1.1.5 CHEMICAL-VAPOUR DEPOSITION:

CVD, or chemical vapour deposition, is a process used to create superior solid materials. Thin films are developed for the semiconductor industry. The procedure, also known as vacuum deposition, is carried out at a lower pressure than the surrounding atmosphere. Chemicals and vapours react on the surface of a substrate to form solids. The CVD technique is used to deposit protective layers such as silicon dioxide, polysilicon, and silicon nitride on the substrate.
As an example,

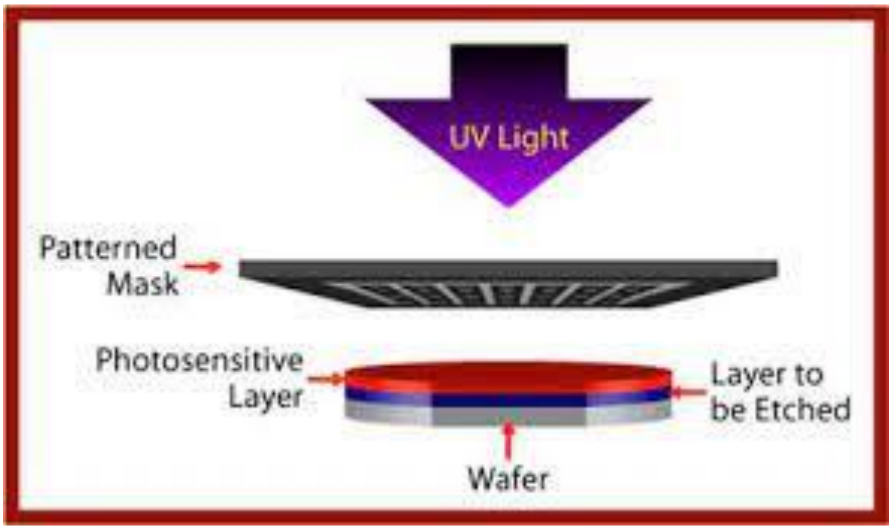
When oxygen on the surface of a silicon wafer reacts with silane gas, a silicon atom molecule, SiO2, or silicon dioxide, is formed.



1.1.6 PHOTOLITHOGRAPHY:

Photolithography, also known as optical lithography, is the use of light to create thin films. It is used to shape thin sheets on a substrate such as silicon wafers. It protects specific locations during a series of manufacturing procedures (deposition, ion implantation, etc.). UV light, X-rays, and severe UV light with varying frequencies are used to introduce mask patterns on the silicon wafer. The most common type of light used to create films is ultraviolet light.

The silicon wafer or substrate is first coated with a photoresist. On the photoresist layer, a photolithographically applied mask pattern covers the silicon wafer.



1.1.7 METALIZATION:

The process of applying a metal layer to a metallic or non-metallic surface is known as "metallization." Silver, zinc, or aluminium might be used as the coating. Aluminium is the metal used in CMOS production to protect the surface from outside environmental elements like water, air, and dust.

1.1.8 PACKAGING:

Packaging is the final step in the production of an integrated circuit. The final silicon wafers or chips, which may contain a plethora of microscopic components, are electrically tested to ensure proper operation. For testing, an automatic probing station is used. It is a reasonably priced testing device with microwave and radio-frequency testing capabilities. All circuits are separated from the default circuits. The functional circuits are routed to headers or packaging.

1.1.9 CONCLUSION:

Wafer preparation precedes the deposition of an oxide layer or protective layer, the introduction of impurities (n- or p-type), the implantation of ions onto the crystal's surface, CVD to produce thin films, photolithography (the use of UV light to create a mask pattern), metal deposition for interconnections, and packaging.

2. BIBLIOGRAPHY:

1. *["Power outage partially halts Toshiba Memory's chip plant"].[Reuters]. [June 21, 2019].*
2. **https://www.javatpoint.com/ic-fabrication-process.**